Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

Problem Statement:

Manual entry of external employee data into ServiceNow is time-consuming, inefficient, and highly susceptible to human error. Furthermore, once user records exist in the system, service desk agents often lack immediate visibility into critical user context—such as a user's department, email, or manager—when resolving Incidents. Navigating away from the Incident form to retrieve this information slows down resolution times and creates a disjointed user experience. There is a need for an automated, scalable solution to reliably bulk-import data and seamlessly surface related user information directly on active task forms.

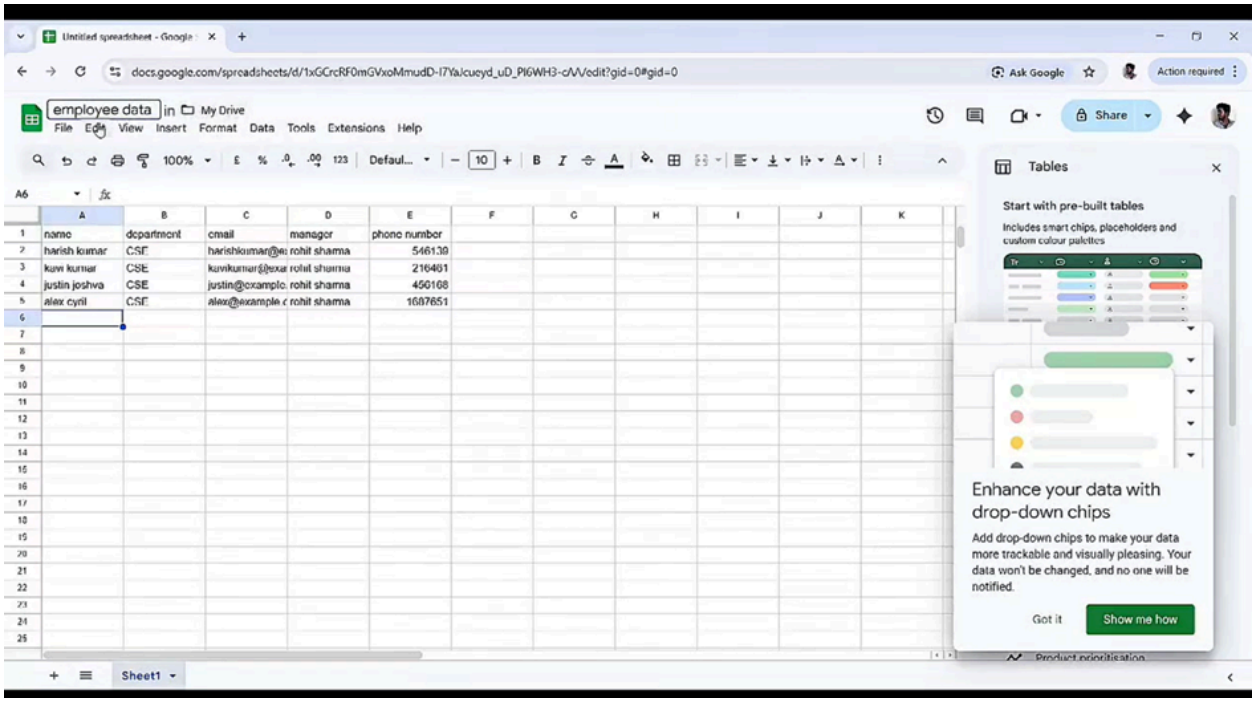
Objective:

To automate the ingestion of external employee data into the ServiceNow `sys_user` table utilizing Import Sets and Transform Maps, ensuring structured and accurate data mapping. Additionally, to enhance the usability of the Incident management process by implementing "Dot-Walking" on the Incident form. This will automatically display related reference data (Department, Email, and Manager) for the Assigned User without requiring custom scripting, ultimately improving data accuracy and reducing manual effort for support agents.

TASK INITIATION:

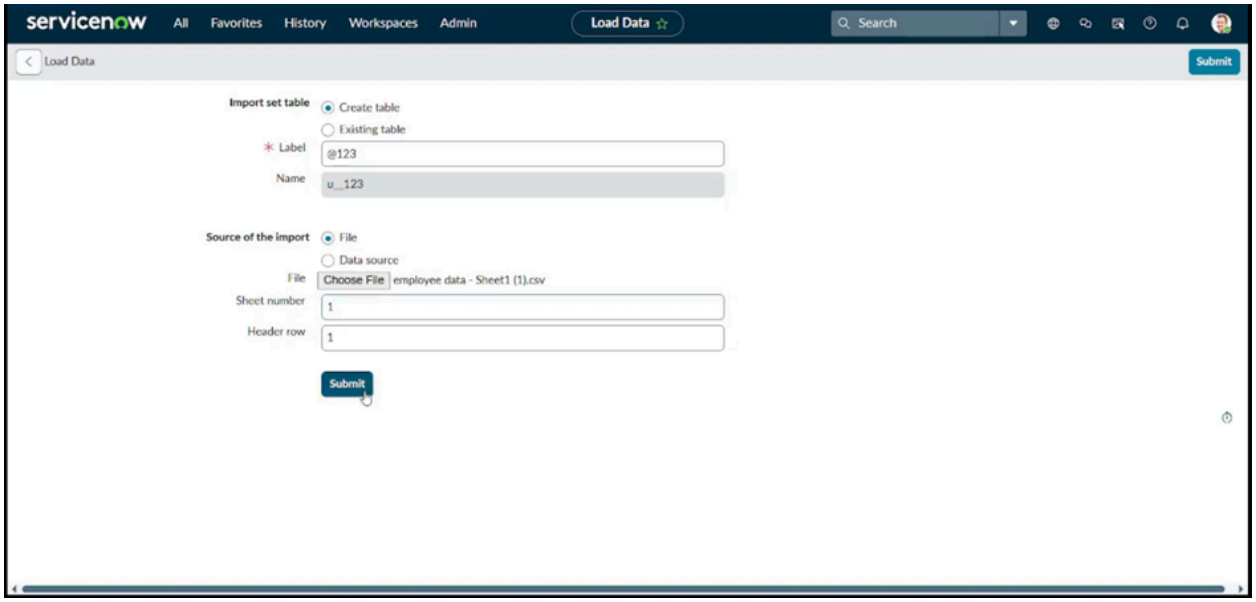
Phase 1: Data Preparation

1. Open Google Sheets (or Excel).
2. Create columns and name the fields as: `Name`, `Email`, `Department`, `Manager`, and `Phone Number`.
3. Enter sample employee records into the sheet.
4. Save the sheet as: `employee data`.
5. Download the file in a supported format (.xlsx, .csv, or .xml).



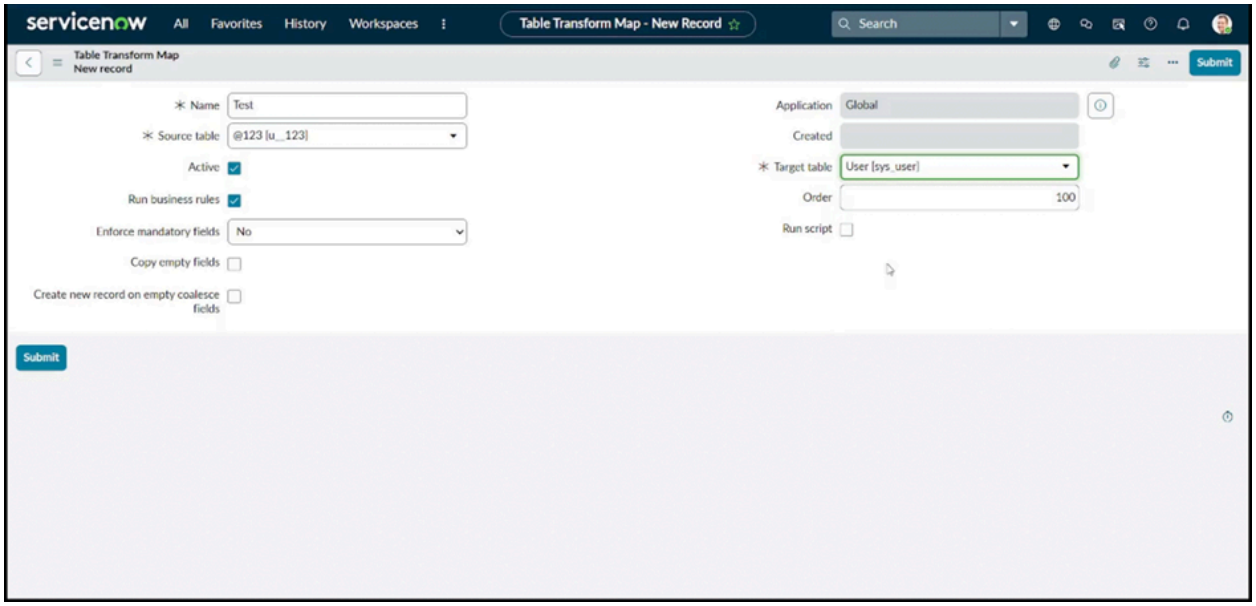
Phase 2: Creating Import Table (Staging Table)

1. Open ServiceNow.
2. Click on **All** and search for **Import Sets**.
3. Select **Load Data** under the System Import Sets module.
4. Select **Create table**.
5. Set the Import set table name as: `import employee`.
6. Choose the file (`employee data`) which was created and downloaded in Phase 1.
7. Submit to load the data into the staging table.



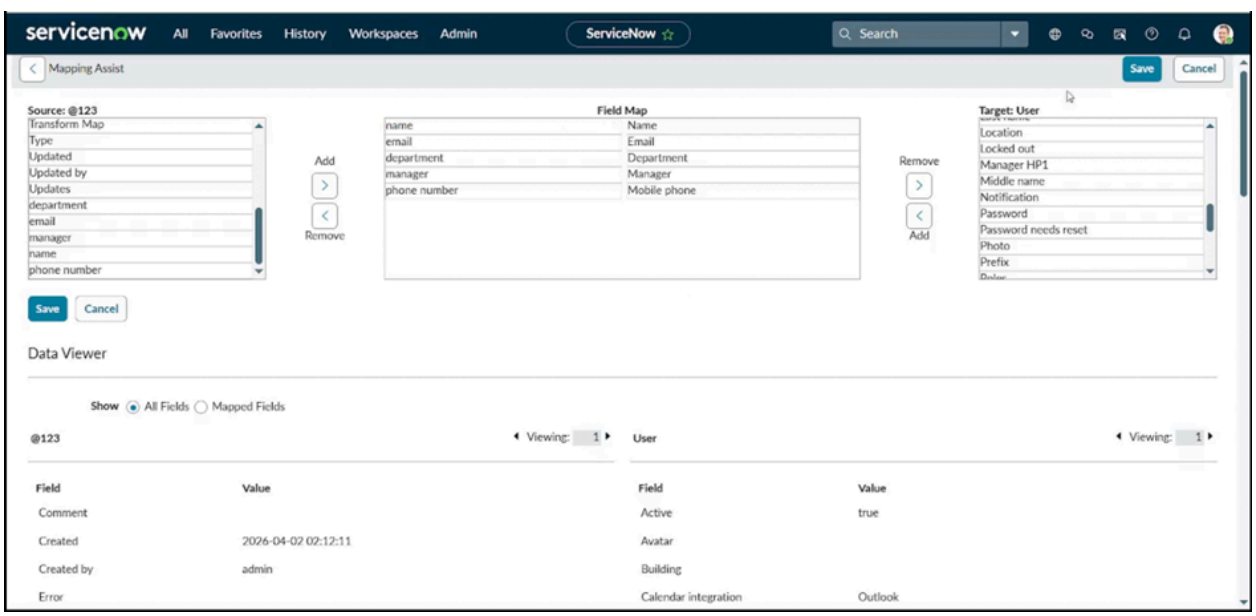
Phase 3: Create Transform Map

1. Navigate back to **All** >> search for **Import Sets**.
2. Select **Create Transform Map** under the Import Sets module.
3. Click on **New**.
4. Fill in the following details to create a new transform map:
    - **Name:** `Test`
    - **Source table:** Select the staging table created earlier (e.g., `@123` / `import employee`).
    - **Target table:** `User [sys_user]`
5. Save the form.



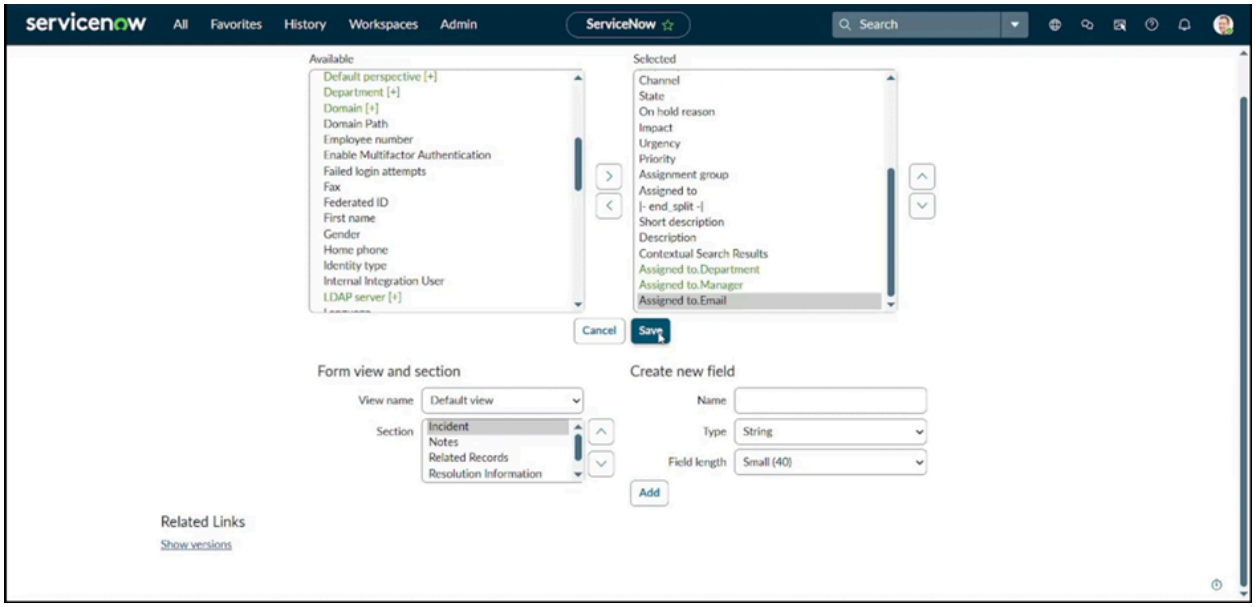
Phase 4: Match the Fields Using Mapping Assist

1. Scroll down to the Related Links on the created Transform Map form.
2. Click on **Auto Map Matching Fields**.
3. Verify that the fields from the staging table are correctly mapped to the target `sys_user` table fields. (Use Mapping Assist manually if any fields do not auto-match).



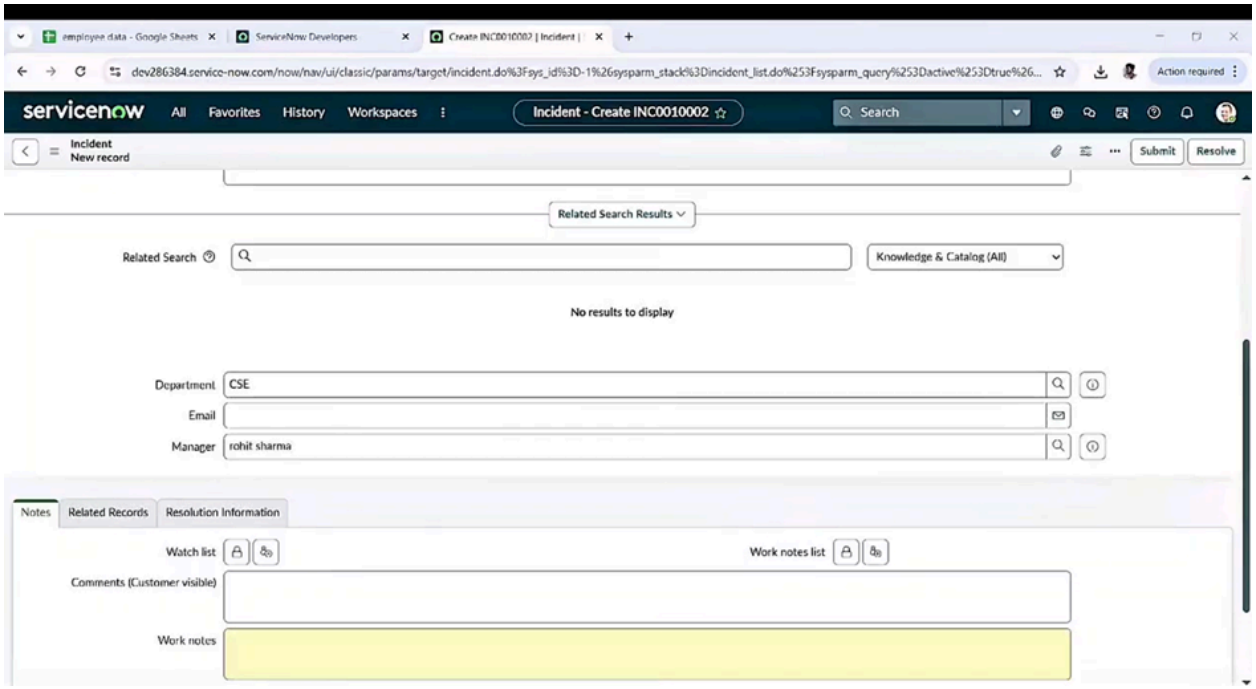
Phase 5: Transform the Data

1. Click on **Transform** in the Related Links.
2. Confirm the selection and click on **Transform** again to execute.
3. Wait for the process to finish and verify that the Data's transform state is marked as **Completed**.



Phase 6: Automating Relationship Mapping (Dot-Walking)

1. Open ServiceNow and navigate to **All** >> **Incident**.
2. Select **Create New** to open a blank Incident form.
3. Notice that fields like *Department*, *Email*, and *Manager* are not visible by default.
4. Right-click the form header (Context Menu), go to **Configure**, and select **Form Layout**.
5. Use dot-walking to retrieve related user data without needing to open the underlying user record:
    - Find **Assigned to** in the Available slushbucket, click the expand icon (+), and select **Assigned to.Department**. Move it to the Selected bucket.
    - Add **Assigned to.Email**.
    - Add **Assigned to.Manager**.
6. Click **Save** to apply the new form layout.



Phase 7: Result Verification

1. Open a new Incident form.
2. Insert a user into the **Assigned to** field (select a user that was imported in the previous phases).
3. Verify that the form automatically populates and updates the read-only *Email*, *Department*, and *Manager* fields based on the selected user's profile.

Conclusion :

This project demonstrates how to efficiently import external employee data into ServiceNow using Import Sets and Transform Maps while maintaining proper data relationships. By mapping reference fields such as department and manager correctly, structured and reliable records are created in the system.

Using dot-walking on the Incident form further enhances usability by automatically displaying related information like department, email, and manager without manually opening the user record. Overall, the solution improves data accuracy, reduces manual effort, and ensures seamless relationship mapping across tables.